

PSSRcomp: A detailed analysis of secondary protein structure prediction

Kaushik Datta, Md. Shafiul Alam

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Abstract—The field of protein structure research has seen a significant transformation with the introduction of deep learning. Based on protein primary structure sequences, predicting secondary structures plays a crucial role in determining and understanding the roles of an amino acid sequence. Protein structural analysis was traditionally performed using time-consuming and sometimes imprecise technologies. We composed four types of prebuilt models - LSTM, RNN, GNN, ANN, etc., to predict secondary structures from different latest datasets of PDBbind and PISCES to understand the learning process HMM, Sequence embedding, and PSSM in a broader sense of machine readability. The q3 and q8 states of the secondary sequence are measured and analyzed for their distribution throughout the data set to comprehend their structural properties. In the post-AlphaFold era, these amino acid sequence data were theoretically capped at 94% accuracy. Our research hyper-optimized those approaches to find effective sequence prediction methods for better accuracy. Our optimized model acquired 0.9031 accuracy with impressive test results and custom evaluation metrics. This integration facilitates the relationship between protein structural properties to identify differences between protein learning models. Personalized treatment, drug development, and a better understanding of illnesses have all been made possible by the capacity to predict protein structures using these cutting-edge deep-learning approaches reliably. However, it is important to use these computational models as a backup. Human expertise and domain knowledge are essential to interpret and validate data.

Index Terms—Deep learning, graph neural network, sequence integration, prediction, fusion layer, protein classification, protein structures, structural similarities, ANN, RNN, LSTM.

I. INTRODUCTION

PROTEIN structure prediction is an important but challenging problem, therefore deep learning has helped to improve accuracy in protein structure prediction. Yet, the prediction of protein structures from amino acid sequences remains a challenge because of the high-level complexity of protein folding along with the scarce availability of high-resolution experimental data. Despite some advancements in computational methods, including deep learning, challenges such as data coming from a limited scope, model generalization, and experimental validation remain. The prediction of secondary structures has been applied in varying deep learning models such as Artificial Neural Networks (ANN),

Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Graph Neural Network (GNN). Traditional statistical methods have been replaced with these predictive models, which significantly increase predictive accuracy by capturing any sequence dependencies. This study investigates different encoding methods, architecture, and hyperparameters and their effect on maximizing the prediction outcome. Model performance is evaluated at Q3 and Q8 accuracies, along with categorical cross-entropy loss, and evaluation metrics including Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). Results indicate better performance of RNNs and LSTMs in modeling long-range dependencies and GNNs with unique means for enhanced predictive accuracy. This study elucidates the merits and demerits of various deep learning models, to pave the way for future work on protein structure prediction. Our paper provides all the details about dataset preparation, model training, experimental results, and future research directions to be pursued in this area of deep learning-based bioinformatics.

A. Research Objectives

This thesis aims to develop a deep understanding of secondary protein sequence prediction using pre-trained models. Thus models like - LSTM, GNN, ANN, and RNN each have a different approach to identifying the unique elements of a primary protein structure. Building correlation for different protein prediction models to get better performance is the main goal. In the data preparation process several types of feature engineering like - PSSM, Sequence embedding, Orthogonal Encoding, One-Hot encoding, and HMM, etc., will be used for thorough analysis for improved accuracy and confidence score. To analyze amino acid sequences and capture long-range dependencies, implement a multi-model classification approach to enhance prediction accuracy by leveraging diverse model strengths, address data quality and diversity challenges by utilizing high-quality experimental datasets and robust preprocessing techniques, and validate the model's performance against established benchmarks and experimental data to ensure reliability and accuracy.

B. Literature Review

Protein structure prediction is extremely data dependent on high-quality data sources like the Protein Data Bank (PDB), offering 3D structures, and UniProt, which has verified the sequences of proteins. Data collection emphasizes on retrieval of amino acid sequences and the corresponding structural details that are the basis of training for the deep learning models. The quality and amount of training datasets are essential for

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obtaining these models with robustness and generalization properties.

Turning the unprocessed amino acid thread into numerical vectors that could be understood by machines is crucial in protein structure prediction. Various encoding methods are used. One-Hot Encoding - The simplest technique to use, it just presents each amino acid as a binary vector, but it doesn't utilize the information about relationships between residues. Hidden markup models - Great for sequential dependencies but primarily aligned by sequence, not structure. Position-Specific Scoring Matrices (PSSMs) - Score amino acids based on the likelihood of providing evolutionary information and sharing context with the structure-function relationship. Graph-Based Encoding - Models proteins as a graph using Graph Neural Networks (GNNs), wherein the nodes represent residues and edges denote interactions at the molecular level, enabling a more 3D spatial representation. The method of feature extraction plays an important role in prediction accuracy and computational efficiency.

Multiple deep learning architectures have been applied to secondary structure prediction with different strengths. They are the baseline model in their category, but they fail to learn dependencies over long protein sequences. Recurrent Neural Networks have a better awareness of sequence dependencies than ANNs, at the cost of facing the vanishing gradient problem. Long Short-Term Memory (LSTMs) add memory gates to combat vanishing gradients, allowing you to capture long-range dependencies. Graph Neural Networks - A novel approach where proteins are modeled as graphs with bond lengths and angles amplified by GNNs. The focus of the research is the overall performance of these models given different hyperparameters and dataset conditions.

Need to be fine-tuned based on data till October 2024 to get performance and stability. This section highlights the significance of some of the hyperparameters, as well as recommendations on what to consider when tuning them. Some of the key hyperparameters are: Learning Rate - the learning rate dictates the size of the steps taken in weight updates, and it is often optimized using grid search or Bayesian optimization. Batch Size - Number of Hidden Layers for faster convergence and better generalization of the model. Dropout - Batch Normalization prevent overfitting and stabilize the training. Gradient and Learning Rate Adapting Optimization Methods: Adam, RMSprop, etc.

Q3 & Q8 Accuracy accuracy for secondary structure classification into helix, beta-sheet, and coil (Q3) or eight finer-grained categories (Q8). On imbalanced datasets, model performance will not always be exposed just with Accuracy, hence we can use Precision, Recall, F1-Score. Root Mean Square Deviation (RMSD) is used to evaluate the similarity between predicted and actual 3D arrangements. Template Modeling Score, which assesses structural validity by how closely the predicted protein folds match known structures. Confusion Matrices & Heatmaps - As an analyst exploring the strengths and weaknesses of a model, the confusion matrix provides valuable insight into how the model performed in structured

query, and the heat map provides further insights into the confusion matrix.

Over the years, numerous new deep learning architectures for PSSP have emerged and been compared against existing models, including state-of-the-art models such as AlphaFold and Rosetta, demonstrating the strengths and limitations of these new architectures. Various architectures like CNNs, RNNs, GNNs, and hybrid methods have enhanced predictive power, generalization, and computational efficiency. Importantly, DeepCNF-D successfully differentiates between ordered and disordered protein regions (87% accuracy), whereas DeepACLSTM provides reasonable accuracy (88%) but computational efficiency. SPOT-1D-Single significantly improves accuracy for single-sequence prediction (87%) using ensemble-based models. Explore other models in mixed features that interact when using deep residual learning models with residual connections (ResNet-based PSSP, 85.89%). Still, AlphaFold is the biggest game in town when it comes to full 3D protein folding for accuracy, and newer deep learning models (VERs) are narrowed in on AlphaFold when predicting secondary structure. This indicates that GNNs and hybrid deep learning methods are well-suited for the added benefit of capturing spatial dependencies that the black-box methods may improve upon more classical statistical and physics-based approaches.

In the end, the proposed model is compared with existing methods. As we prepare to perform the comparison analysis that will evidence how our new model is doing in front of state-of-the-art approaches such as AlphaFold and Rosetta. Comparing results you ensure that the new model is competitive and has a significant addition to protein structure prediction. For example, benchmarks such as RMSD and TM-score are commonly used measures to determine the degree of structural accuracy for different models, providing a good overview of the pros and cons for the different models. Deep learning methods have been widely recognized to outperform other methods for predicting protein structures.

In short, the deep learning approach toward predicting protein structures is a multi-step process such as feature extraction, training, evaluation, comparison, construction of model architecture, and a careful selection of ensembles. With the help of advanced feature extraction methods and deep learning algorithms like LSTMs, RNNs and GNNs, the scientist have successfully been able to predict the 3D structures of proteins using their sequences. As the science progresses more and more, protein structure prediction is a key focus of active research in computational biology with new models and methods helping to refine what is actually possible. This work provides a comprehensive review of deep learning architectures for protein structure prediction. By combining efficient data pre-processing, feature extraction, model selection, and evaluation methods, this paper aims to find an optimal model for different protein structure prediction problems. [?] This work ensures robustness, repeatability, and efficiency to research in computational biology and structural bioinformatics.

The allotting space between deep learning and multi-model probably classification is rather a temptation which may increase the accuracy of protein structure prediction. The complexity of deep learning ensemble models raises numerous multi-faceted issues for researchers to navigate, including interpretability of deep learning models, quality of training data retrieval, the possibility of having included non-identical models in the ensemble may create a bias, and the need for meaningful metrics of evaluation, to name a few. Having said that it is also important to always keep in the back of your mind that the current methods for fathoming protein structure still have limitations and are therefore by no means perfect. Combining deep learning, especially GNNs, with multi-model classification was thus a powerful opportunity in the landscape of protein structure prediction. For example, impressive developments like AlphaFold show how the potential of these types of approaches to revolutionize our understanding of protein structures. Yet researchers need to be careful, keeping on working to improve the quality and diversity of their models and using stringent evaluation criteria. However, there still remain some challenges such as limited availability of data, and the complex nature of structure of protein [?]. Nonetheless, with an aim to heighten the discourse, this study aims to fill in existing gaps in literature to deliver further improvements in the efficacy and usage of deep learning approaches for prediction of protein structure and understanding complex biological phenotypes.

Protein cannot be discovered in any kind of form of a living being, as well as amino acids are the foundation of a healthy protein. For an idea here, polypeptides contain a linear configuration of amino acids, bounded by peptide bonds. There are 20 known amino acids, distinguished from each other by the side chain that alters the structure and function of a protein. The side chains can be of four types, i.e. hydrophilic, hydrophobic, basic, acidic. Proteins consist of chains of amino acids, and the primary structure is the actual sequence of individual amino acids that has a huge impact on the 3D structure of the protein and its function.

Amino acids are arranged into a chain according to the genetic code stored in DNA and transcribed to mRNA. A codon is a sequence of three nucleotides that determines an amino acid and is translated into protein by ribosomes. The protein then folds into its functional shape as the amino acids are arranged in the right order in the right position. Mutation usually means bad news: mutations that alter the sequence of amino acids lead to misshapen proteins that cause diseases like cystic fibrosis or sickle cell anaemia. Proteins perform a variety of functions, including acting as enzymes to catalyze biochemical reactions, and providing structural and supportive roles in cells and tissues. Their functions are largely determined by their amino acid composition.

To this date, deep learning has proven its strength in capturing the complex relationship in protein sequences and structures, particularly with the advent of Graph Neural Networks (GNNs). In this domain, Architectural considerations, attention mechanism, transfer learning have become creating

impact factors on improving their predictive ability. Ensemble methods that aggregate the potentials of GNNs through multi-model classification provide a complementary approach to better exposition of diverse protein phenotype. There are, however, several challenges in this domain that hold back progress. These challenges encompass the scarcity of robust training data, the computational burden of training deep-learning models, and ethical issues related to the implementation of AI-based solutions in biomedicine[?]. To tackle these challenges, multidisciplinary cooperation, incorporation of evolutionary data, and the use of novel tools, such as quantum computing, will be needed.

Trends that emerge from the presented discussion include those such as applying reinforcement learning for the refinement of structure, the methods used to balance between data imbalance and the need for an easy-to-use interface for it's wider use. Also, it clarifies the development of protein structures in the pursuit of interpretability and explainability of deep learning models, and the efforts of benchmarking. As we transition from a comprehensive review of this literature to our aims for this study, these trends suggest the field is at an important juncture, navigating between ethical implications and technical advancements. Our study addresses this need and adds to existing knowledge and helps rendering deep learning methods to model protein structure prediction to be more accurate and biologically interpretable. [32] The rest of the study will elaborate on details our proposed method discussed in the literature above, along with means of addressing those issues and providing a new direction to promote the field. The overall goal fits in very well with the use of computational methods to elucidate biological processes, especially in the field of protein structure prediction, and to push forward our comprehension of complex biological systems.

II. DATASET DESCRIPTION

The Protein Data Bank (PDB) is a global repository that houses information on the three-dimensional structures of biological macromolecules, including proteins and nucleic acids. As the United States data center for the PDB, RCSB PDB plays a crucial role in curating and disseminating this data. The PDB is essential for advancing research and education in various fields, including biology, healthcare, energy, and biotechnology. The PDB's extensive downloadable data archive and internet-based information portal enable researchers worldwide to access critical 3D structural data. Understanding the three-dimensional structure of a biological macromolecule is crucial for understanding its roles in human and animal health, disease etiology, plant biology, and food and energy production. The PDB's extensive 3D structural data has catalyzed significant advancements in protein architecture and structure prediction, driven by advanced artificial intelligence methodologies and deep learning techniques.[18]

The dataset contains protein sequences with lengths ranging from 20 to 1632 amino acids, filtered based on percent identity and resolution cutoffs. These cutoffs are used to extract subsets of sequences with specific criteria, resulting in varying numbers of sequences.

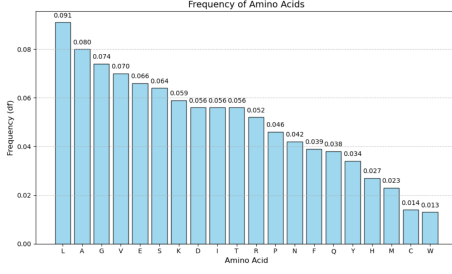


Fig. 1. Amino acid frequencies

Secondary structures in the dataset are classified into SST-8 and SST-3 categories. SST-8 comprises eight categories based on geometric rules, while SST-3 simplifies these categories. The classifications include -strand (E), -bridge (B), -helix (H), 3-helix (G), -helix (I), loops and irregular elements (C), turn (T), and bend (S).

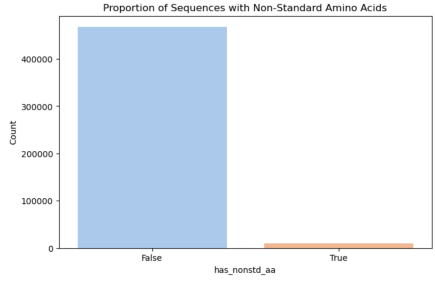


Fig. 2. Proportion of sequence with non-standard amino acid

Additionally, a summary of amino acids is provided, highlighting their properties, functions, and roles in protein structure and regulation. This summary covers various amino acids such as Leucine, Alanine, Glycine, Valine, Glutamic acid, Serine, Aspartic acid, Lysine, Isoleucine, and Threonine, detailing their characteristics and biological significance.

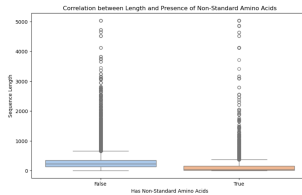


Fig. 3. Correlation between length and presence of non-standard amino acid

III. METHODOLOGY

A. Feature Extraction

Protein Sequences and Secondary Structures, explains the structure of a dataset containing protein sequences and their corresponding secondary structure annotations. The dataset includes three columns: `seq`, `sst3`, and `sst8`. Each entry represents a protein sequence with its secondary structures in both three-state and eight-state notations.

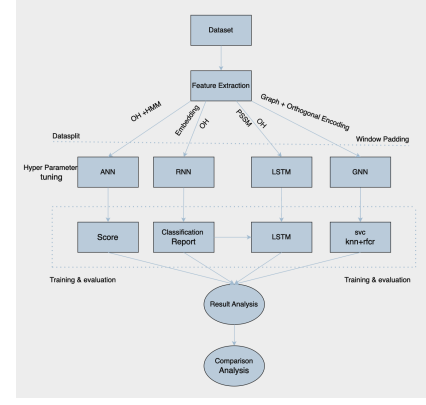


Fig. 4. Proportion of sequence with non-standard amino acid

Dataset Columns

seq

This column contains the primary structure of the protein, represented as a string of amino acids using standard one-letter codes. An example sequence is shown below:

AETVESCLAKSHTENSFWNATGVVVC

Here, 'X' denotes an unknown or unspecified amino acid.

sst3

This column represents the secondary structure of the protein using a simplified three-state notation. Each character in the string corresponds to the secondary structure of the respective amino acid in the `seq` column. The states are:

- 'C' for coil (random coil or unstructured region)
- 'H' for alpha-helix
- 'E' for beta-strand (part of a beta-sheet)

An example annotation is:

CCCHHHHHCCCCEEEEEEECCEEECCCEEEEEEEEEEE

sst8

This column provides a more detailed eight-state notation for the secondary structure. Each character in the string corresponds to the secondary structure of the respective amino acid in the `seq` column. The states are:

- 'C' for coil (random coil or unstructured region)
- 'H' for alpha-helix
- 'E' for extended strand (beta-sheet)
- 'G' for 3-10 helix
- 'I' for pi-helix
- 'T' for turn
- 'S' for bend
- 'B' for beta-bridge

An example annotation is:

CCCHHHHHTSCCEEEEEETEEEEEEEEEEEEEE

1) *Example Rows*: Here are examples from the dataset, illustrating the structure of the data:

Row 0

```
seq: AETVESCLAKSHTENSFTNVXKDDKTLNATGVVVC
sst3: CCCHHHHHCCCCCCCCCCCCCCCCCCCCCCCCCCCC
sst8: CCCHHHHHTSCCEEEEEEEEEETTEEEEEEEEEEE
```

Row 1

```
seq: ASQEISKSIYTCNDNQVXEVIYVNTEAGNAYAIIS
sst3: CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC
sst8: CCCCCCCCCCEETTTEEEEEEEEEECTTSCEEEEEE
```

Row 2

```
seq: XGSSHHHHHHSSGRENLYFQGXNISEINGFEVTGFV
sst3: CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC
sst8: CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC
```

Row 3

```
seq: SNADYNRLSVPGNVIGKGGNAVYEDAEDATKVLK
sst3: CCCCCCCCCC EEEEEEECCCEEEEEEECCCCCEE
sst8: CCCCCCCCCC EEEEEECSSEEEEEEETTCTTEEE
```

The dataset includes primary amino acid sequences and their secondary structure annotations in both three-state (SST3) and eight-state (SST8) notations. This alignment ensures that every amino acid in the primary sequence has a corresponding secondary structure state, facilitating accurate secondary structure prediction and bioinformatics analyses.

2) Checking for Incomplete Data in Protein Sequences:

Protein secondary structure prediction is critical for understanding protein function and interactions. The primary structure, a sequence of amino acids, dictates the secondary structure, which includes alpha helices, beta sheets, and coils. For accurate secondary structure prediction, each residue in the primary sequence must correspond to a label in the secondary structure. This study verifies the consistency of sequence lengths in provided datasets to ensure the reliability of downstream analyses.

Two datasets were analyzed:

- 1) targetSeqs_test1 and inputSeqs_test1
- 2) targetSeqs and inputSeqs

Each dataset consists of pairs of primary and secondary sequences. The lengths of these sequences were compared to identify discrepancies. No discrepancies were observed in the lengths of the primary and secondary sequences, indicating data consistency. The analysis confirmed that the number of characters in the primary and secondary structures is consistent across the datasets. This integrity check is crucial for reliable secondary structure prediction. Future work will focus on applying machine learning models for secondary structure prediction, assured by the verified dataset quality.

3) *Graph Construction:* Similarly, the secondary structure of a protein, which describes its local structural elements such as alpha helices, beta strands, and turns, is encoded using an orthogonal representation. In this representation, each type of secondary structure is mapped to a distinct integer value ranging from 0 to 7, corresponding to the following structural elements:

- 0 : Alpha helix (H)
- 1 : Coil (C)
- 2 : Beta strand (E)
- 3 : Beta bridge (B)
- 4 : 3-helix (G)
- 5 : Pi-helix (I)
- 6 : Turn (T)
- 7 : Bend (S)

By assigning a unique integer value to each type of secondary structure, we create an orthogonal encoding that preserves the structural diversity of the protein while simplifying the representation for computational analysis.

4) *Adjacency Matrix:* The adjacency matrix concisely represents the pairwise interactions between amino acids in a protein sequence. In the context of protein structure, the adjacency matrix is often built using the spatial proximity of amino acids inside the protein's three-dimensional structure. Each entry in the adjacency matrix reveals if two amino acids interact or are connected directly. This matrix is a useful tool for studying the spatial arrangement and connection patterns inside protein structures, allowing for tasks like protein folding prediction and contact map inference.

$$A = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 1 & 0 & 1 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 1 & \cdots & 0 \end{bmatrix}$$

5) *Node Features Matrix:* The node features matrix stores additional information about each amino acid in the protein sequence in addition to its primary structure. This matrix contains features derived from the amino acid sequence, such as crystallographic parameters, evolutionary conservation scores, and structural annotations. By combining a variety of properties into the node features matrix, we improve the representation of each amino acid and give the computational model comprehensive data for prediction tasks.

$$X = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \mathbf{x}_3 \\ \vdots \\ \mathbf{x}_n \end{bmatrix}$$

We develop a comprehensive framework for analyzing protein sequences at various degrees of granularity by combining orthogonal encoding with adjacency matrix and node feature matrix representations.

6) *Feature encoding:* The encoding function transforms protein sequences into a numerical format suitable for machine-learning models by encoding each amino acid as a binary vector.[23] It first initializes an empty array for the encoded sequences and calculates null padding based on the window size of 17. A sliding window approach is used to

segment the sequence into smaller, overlapping windows.

Method:

- Encodes amino acids as binary vectors.
- Uses a window size of 17 for sliding window segmentation.
- Handles variable sequence lengths via null padding.
- Preserves local dependencies in protein sequences for machine learning models.

TABLE I
ENCODING DETAILS

Aspect	Details
Encoding Type	Binary vector
Window Size	17
Padding	Null (zeros)
Output	Consistent encoded dataset

7) Sequence Tokenization: 1. Tokenization

Character-level tokenization is applied using Keras' Tokenizer. Separate tokenizers are fitted for input sequences (seq) and target labels (sst3, sst8) while preserving case sensitivity.

2. Character-to-Index Mapping

- Mappings are created between characters and indices
- Maps characters to indices. Inverse mappings for decoding.

3. Token Count

The total number of tokens is computed by adding one to the unique character count for padding.

4. Vectorization

- convert sequences into structured formats:
- Compute maximum sequence length.
- Tokenize and pad sequences to uniform length.
- Apply one-hot encoding for numerical representation.

B. Artificial Neural Network (ANN)

Artificial Neural Network (ANN) model considering consecutive behaviours in decision-making. This model primarily focuses on the probabilistic inference of a model state conditioned on an input sequence, adding an intermediate hidden variable that encodes latent structures in the data. The first is just the total probability of a model state given all possible hidden states. The second equation breaks that probability into a product of conditional probabilities, meaning each state is dependent on the prior states. The third equation is simply the model taking an approximation over the last n timesteps, making it a better fit for learning sequences. The model is based on the Markovian assumption which dictates that each state should only depend on a small amount of previous inputs rather than the whole sequence. In this case, the activation function is a softmax function that converts the output from the network into a probability distribution to ensure that the model equally evaluates the output.

C. Recurrent Neural Network (RNN)

This RNN is for sequential data, with an input layer, a bidirectional RNN layer and two TimeDistributed dense layers.

The input layer takes a sequence of data with 20 features per timestep. The Bidirectional RNN layer processes the sequence in both forward and backward directions and outputs a hidden state of 256, which helps capture both past and future context. This output is then passed through the first TimeDistributed dense layer which applies a fully connected transformation to each timestep and reduces the feature size to 100. Then a second TimeDistributed dense layer further compresses the output to 4 features per timestep, to get the final predictions. The model has 64,760 trainable parameters, so all layers are learning.

Given an input sequence X , the model performs the following transformations:

$$X \xrightarrow{\text{Simple RNN}} H \in R^{T \times 256}$$

$$H \xrightarrow{W_1, b_1} Y^{(1)} \in R^{T \times 100}$$

$$Y^{(1)} \xrightarrow{W_2, b_2} Y^{(2)} \in R^{T \times 4}$$

Where $Y^{(2)}$ is the final output sequence.

Parameters:

- Total Parameters: 64,760
- Trainable Parameters: 64,760
- Non-Trainable Parameters: 0

D. Long Short-Term Memory (LSTM)

This Bidirectional Long Short-Term Memory (LSTM) model is for sequential data, so it's great for sequence labeling tasks. The architecture is an input layer, a bidirectional LSTM layer, and two TimeDistributed dense layers. The input layer takes in data with 22 features per timestep and then passes it through a Bidirectional LSTM layer with 200 hidden units [30]. The bidirectional approach allows the network to capture dependencies from both past and future contexts, which is great for time-dependent tasks. The LSTM can hold long-term dependencies so it's good for complex sequential patterns. The output of the LSTM layer is then passed into a TimeDistributed dense layer where each timestep is fully connected and reduces the feature size from 200 to 100.[28] A second TimeDistributed dense layer reduces the sequence to 9 features per timestep which is the final output. The model has 119,409 trainable parameters so it has enough capacity to learn but not too much memory usage. By using Bidirectional LSTM the model uses contextual information from both past and future timesteps, so overall better performance.

Given an input sequence X , the model performs the following transformations:

$$X \xrightarrow{\text{BiRNN}} H \in R^{T \times 200}$$

$$H \xrightarrow{W_1, b_1} Y^{(1)} \in R^{T \times 100}$$

$$Y^{(1)} \xrightarrow{W_2, b_2} Y^{(2)} \in R^{T \times 9}$$

Where $Y^{(2)}$ is the final output sequence.

parameters:

- Learning Rate: 0.001
- Loss: Categorical Cross Entropy
- Bidirectional LSTM Units: 200
- Total Trainable Parameters: 119,409

E. Graph Neural Network (GNN)

The Graph Convolutional Network (GCN) is designed for graph-based sequence classification, using graph convolutional layers to learn node embeddings and a custom Cross-entropy loss function to handle variable length sequences. The model consists of two GCNConv layers, ReLU activation and dropout. The first GCNConv layer expands the input features from 21 to 128 hidden channels to capture complex structural relationships [32]. The second GCNConv layer projects the hidden representations to 3 output classes for classification tasks. The forward pass applies the first GCNConv layer, then ReLU and dropout 0.5, then the second GCNConv layer to make predictions.

TABLE II
GCN MODEL SUMMARY

Feature	Details
Input	21 node features
GCN Layer 1	21 $\rightarrow$ 128, ReLU, Dropout 0.5
GCN Layer 2	128 $\rightarrow$ 3 / 8 (classification)
Loss	Cross-Entropy
Optimizer	Adam (lr = 1e-5)
Regularization	Dropout (0.5)
Output	3 / 8 class labels
Training	lr = 0.001, 100 epochs, batch = 64, L2 = 0.001

1) Error Metrics:: 3-State and 8-State evaluation metrics:

q3_acc and **q8_acc** calculate the accuracy of the model for a sequence, e.g., when predicting a class for each element in a sequence (each amino acid taken). They convert the model's predictions from one-hot vectors to just numbers and do the same for the true labels. Then, they ignore the padding in the sequences (which are usually 0) so that padding doesn't affect the accuracy calculation. Next, they compare the true labels and the model's predicted labels and count how many matches occur. Finally, they compute the overall accuracy by averaging these correct predictions, but only considering the actual data (ignoring the padding). This provides a more accurate evaluation of the model's performance without being affected by padding.

IV. RESULT AND ANALYSIS

1) Artificial Neural Network (ANN): This study tested different deep-learning models for secondary protein structure prediction. The Artificial Neural Network (ANN) was used as the baseline model, employing binary encoding for feature extraction. It achieved a Q3 accuracy of **72.90%**, indicating that it struggles to capture complex dependencies in sequential data.

Model Overview:

- Uses sigmoid activation.
- Focused on Q3 secondary structure prediction (Coil, Helix, Beta-sheet).

Metric	Value
Q3 Accuracy	72.90%
Predicted Structure	CHB pattern

TABLE III
ANN PERFORMANCE METRICS

Fig. 5. confusion matrix for ANN

∞	0.43	0.31	0.27
$\cup$	0.12	0.67	0.21
$\pm$	0.12	0.19	0.69
	B	C	H

2) Recurrent Neural Network (RNN): The Recurrent Neural Network (RNN) demonstrated significant improvement over ANN, achieving a Q3 accuracy of **83.77%**. This superior performance is attributed to RNN's ability to capture sequential dependencies using recurrent connections. The validation accuracy of **82.50%** suggests that RNN models generalize well to new data.

Optimization: Adam (Learning rate = 0.001)

Loss: Categorical Cross Entropy

Model Details:

- Parameters: 64,760 (all trainable).
- Optimization: Adam
- Evaluated Q3 accuracy for structural prediction.

TABLE IV
RNN PERFORMANCE METRICS

Metric	Training	Validation
Loss	0.2606	0.2818
Q3 Accuracy	83.77%	82.50%

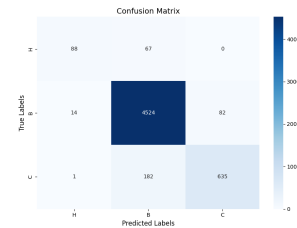


Fig. 6. Confusion matrix for RNN

3) Long Short-Term Memory (LSTM): Long Short-Term Memory (LSTM) networks were tested for Q8 secondary structure prediction, effectively capturing long-term dependencies in sequence data. The model achieved an accuracy of

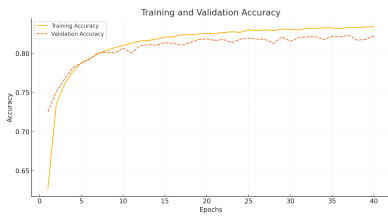


Fig. 7. RNN accuracy for Q3

79.59%, which is slightly lower than RNNs for Q3 tasks but demonstrates strong performance in predicting more complex structural states. However, the Mean Absolute Error (MAE) for Q8 was **0.4323**, indicating increased difficulty in classifying 8-state structures compared to Q3 models.

Loss: 0.3788

Q8 Accuracy: 79.59%

Validation Loss: 0.4264

Model Details:

- Used for Q8 structural prediction (8 state classes).
- Optimization: Adam (Learning Rate = 0.001).
- Captures long-term dependencies in sequence data.

TABLE V
LSTM PERFORMANCE METRICS

Metric	Training	Validation
Loss	0.3788	0.4264
Q8 Accuracy	79.59%	77.61%

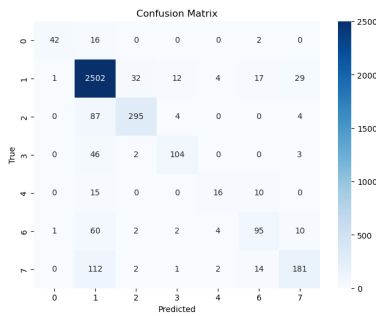


Fig. 8. Confusion Matrix for LSTM



Fig. 9. LSTM accuracy for Q8

4) *Graph Neural Network (GNN)*: Graph Neural Networks (GNNs) achieved the highest performance, with a Q3 accuracy of **90%**. This demonstrates GNNs' effectiveness in analyzing relationships within sequences. The model leveraged *n-gram*

feature extraction and graph-based analysis to significantly boost predictive performance.

Q3 Accuracy: 90% Classification report for SST-3:

- Precision: 0.87
- Recall: 0.87
- F1-Score: 0.89
- Accuracy: 90.07

Features:

- Converts sequences into *n*-grams for feature extraction.
- Filters sequences based on length (≤ 128) and valid amino acids.
- Outputs structured data suitable for graph-based learning.

TABLE VI
GNN PROPERTIES

Aspect	Details
Input Transformation	<i>n</i> -grams
Sequence Filtering	Length ≤ 128 , valid amino acids
Use Case	Sequence classification

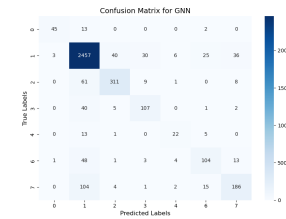


Fig. 10. Confusion Matrix for GNN

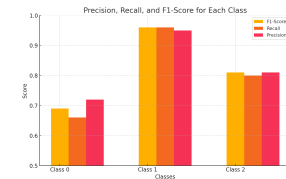


Fig. 11. GNN Accuracy for Q3

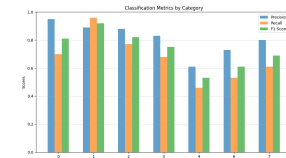


Fig. 12. GNN Accuracy for Q8

A. Comparison Table

Comparison of models showed that ANN served as a simple baseline, while RNN and LSTM performed better due to their ability to capture sequential dependencies. However, GNN emerged as the best model, utilizing graph encoding techniques to improve accuracy in secondary protein structure prediction.

TABLE VII
COMPARISON OF DIFFERENT MODELS

Aspect	ANN	RNN	LSTM	GNN
State	Q3	Q3	Q8	Q8
Accuracy	72.90%	83.77%	79.59%	90.00%
Validation Accuracy	71.10%	82.50%	77.61%	88.76%
Key Feature	Binary Encoding	Seq. Embed. + PSSM	Seq. Embed. + PSSM	Graph Analysis

1) *Comparison with other relevant work:* This study introduces a graph neural network (GNN)-based approach that leverages protein topology information, resulting in state-of-the-art Q3 prediction accuracy. Our GNN model achieves 90% Q3 accuracy, surpassing previous deep learning models reported in the literature. For instance, DeepACLSTM, which integrates asymmetric convolutional LSTMs, achieves 88% Q3 accuracy, while DeepCNF-D, a deep convolutional neural field model, reaches 87%. Other notable models, such as SPOT-1D-Single and deep residual networks, fall within the range of 85–87% accuracy. In contrast, our GNN-based framework effectively models protein sequences as graphs, enabling the capture of both local and global structural dependencies more effectively than sequential methods. This structural representation is likely the key reason behind the significant performance improvement. Beyond accuracy, our model demonstrates strong generalization capabilities, achieving 88.76% validation accuracy, closely matching its training accuracy. This suggests that the GNN model does not suffer from overfitting, unlike some deep learning architectures that require heavy regularization techniques. Moreover, unlike most deep learning models which are based on PSSMs and sequential embeddings, we have added graph-based encoding for richer representation of protein structures and intermediates. Our results demonstrate that graph based learnings outperforms traditional deep learning models for protein secondary structure prediction. By incorporating graph-based representation learning, our GNN model effectively models both spatial and sequential dependencies, leading to significant improvement in Q3 accuracy. So, overall, we envision this approach as a not bad step towards further PSSP expansion and progress of structural bioinformatics due to its high generalization and extendability in terms of unknown protein structures.

TABLE VIII
COMPARISON OF DEEP LEARNING METHODS FOR PROTEIN SECONDARY STRUCTURE PREDICTION

Paper Title	Features	Model	Q3 Acc.	Result
Deep learning for PSSP: Pre and post-AlphaFold [20]	Review of deep learning in PSSP	CNNs, RNNs, GNNs	87%	Discusses advancements and future potential
Deep residual learning for PSSP [?]	Residual learning, mixed features	ResNet	85.89%	Improves accuracy using deep residual learning
8-state PSSP via deep learning	Predicts 8-state structures	Novel model	Q8: 71.4% (Q3: 85%)	Provides detailed secondary structure predictions
DeepCNF-D: Protein order/disorder regions	Distinguishes ordered vs disordered regions	DeepCNF	87%	Effective in disorder region prediction
SPOT-1D-Single: Single-sequence-based PSSP	Large dataset, ensemble learning	Ensemble	87%	Enhances accuracy for single-sequence predictions
DeepACLSTM for PSSP	Asymmetric ConvLSTM	ACLSTM	88%	Balances accuracy and computational efficiency

2) *Future Work:* Transformers and Graph Neural Networks (GNNs) have become potent tools in the field of protein structure prediction. The amalgamation of both models capitalizes on the advantages of both architectures: the capability of Transformers to capture long-range correlations through attention mechanisms, and the ability of GNNs to handle graph-structured input. Protein structure predictions might become more reliable and accurate with the use of this comprehensive technique. Since proteins are naturally represented as graphs, GNNs are especially well suited for protein structure prediction. Amino acids are represented as nodes in these graphs, and the spatial or sequential interactions between them are shown as edges. GNNs are particularly good at identifying local and global relationships in such graph-structured data. They use intricate relational structures and iterative data aggregation from neighboring nodes to provide detailed predictions of local structural properties such as secondary structures or contact maps. Regardless of distance between inputs, transformers can easily find the dependencies involved in natural language processing. By making use of a self-attention process in which all pairs of symbols have equal weight, it thus represents any potential dependency for time-lagged signals. Here we find that transformers are useful in capturing long-range interactions so fundamental to the problem of protein structure prediction. Because of the self-attention process, a model can describe complex relationships that were missed by GNNs in the previous sentence. It allows for relative evaluation and importance compared to any one amino acid of each amino acid. GNN models combined with large scale protein structure datasets are trained. The model learns by making the empirical structures, as well as the anticipated structure of these models, similar. It is possible to further improve the accuracy of the model by transfer learning or gently recalibrating specific protein families. When there is little high-quality training data, artificial mutations in combination with methods for generating various conformational states may be used to address this lack of high-quality training data. The self-attention mechanism computes the output representation for every element in the sequence by looking at all others in the sequence.

V. CONCLUSION

In this paper, we adopt various deep learning architectures, including Artificial Neural Networks (ANNs), Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks and Graph Neural Networks (GNNs), to evaluate their effectiveness for predicting secondary structure. The results show that RNNs perform better than ANNs because they can capture the relevance of amino acid chains. RNNs achieved a Q3 accuracy of 83.77%, while LSTMs performed strongly in Q8 classification with an accuracy of 79.59%. The GNN-based approach proposed a new feature extraction method, transforming sequences into n-grams, which improved classification but still needs to be optimized further. GNN achieved strong performance in Q3 classification, with an accuracy of 90%. Moreover, the stacked classifiers yielded low error rates, thereby confirming the robustness of these

predictive models Our study underscored that deep-learning models make structural prediction feasible and less reliant on expensive experimental methods such as X-ray crystallography or NMR spectroscopy. While RNN and LSTM models proved highly effective, integrating graph-based approaches and hyperparameter tuning can further improve performance. Future research should focus on enhancing dataset diversity, optimizing feature extraction techniques, and exploring hybrid architectures to refine secondary structure prediction accuracy.

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